# QUBO Formulation: Score-Based Bayesian Network Structure Learning with Candidate Parent Sets

## Score-Based Bayesian Network Structure Learning with Candidate Parent Sets

**Problem Statement.** We consider the problem of learning the structure of a Bayesian network over a fixed set of $N$ nodes $\mathcal{N} = \{0, 1, \ldots, N-1\}$ with a predefined topological order. For each node $i \in \mathcal{N}$, a finite set of candidate parent sets $\mathcal{C}_i$ is provided, where each candidate set $j \in \mathcal{C}_i$ contains only nodes with indices strictly less than $i$, thereby guaranteeing a directed acyclic graph. Each candidate set $j$ is associated with a score $s_{i,j} \in \mathbb{R}$ and a cardinality $d_{i,j} \in \mathbb{Z}_{\geq 0}$ representing the number of directed edges it introduces. The objective is to select exactly one candidate parent set for every node such that the total number of directed edges in the resulting network equals a prescribed integer $K$, while maximizing the sum of the scores of the selected sets.

**Decision Variables.** For each node $i \in \mathcal{N}$ and each candidate parent set $j \in \mathcal{C}_i$, we introduce a binary decision variable $x_{i,j} \in \{0, 1\}$. The variable $x_{i,j}$ takes the value 1 if and only if candidate parent set $j$ is selected for node $i$, and 0 otherwise. The full decision vector is $\mathbf{x} = (x_{i,j})_{i \in \mathcal{N}, j \in \mathcal{C}_i}$.

**QUBO Derivation. Step 1 — Objective.** We convert the maximization problem to a minimization problem by negating the objective function:

$$\min_{\mathbf{x}} \left( -\sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} s_{i,j} x_{i,j} \right). \tag{1}$$

**Step 2 — Constraints.** The problem is subject to two equality constraints:

$$\sum_{j \in \mathcal{C}_i} x_{i,j} = 1, \quad \forall i \in \mathcal{N}, \tag{2}$$

$$\sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j} x_{i,j} = K. \tag{3}$$

**Step 3 — Penalty Terms.** We incorporate these constraints into the objective using quadratic penalty terms with weight $\lambda > 0$:

$$H_{\text{pen}}(\mathbf{x}) = \lambda \sum_{i \in \mathcal{N}} \left( \sum_{j \in \mathcal{C}_i} x_{i,j} - 1 \right)^2 + \lambda \left( \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j} x_{i,j} - K \right)^2 . \tag{4}$$

Expanding the first term using the idempotent property $x_{i,j}^2 = x_{i,j}$ for binary variables:

$$\begin{aligned} \lambda \sum_{i \in \mathcal{N}} \left( \sum_{j \in \mathcal{C}_i} x_{i,j} - 1 \right)^2 &= \lambda \sum_{i \in \mathcal{N}} \left( \sum_{j \in \mathcal{C}_i} x_{i,j}^2 + 1 - 2 \sum_{j \in \mathcal{C}_i} x_{i,j} \right) \\ &= \lambda \sum_{i \in \mathcal{N}} \left( \sum_{j \in \mathcal{C}_i} x_{i,j} + 1 - 2 \sum_{j \in \mathcal{C}_i} x_{i,j} \right) \\ &= \lambda \sum_{i \in \mathcal{N}} \left( 1 - \sum_{j \in \mathcal{C}_i} x_{i,j} \right) . \end{aligned} \tag{5}$$

Expanding the second term:

$$\lambda \left( \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j} x_{i,j} - K \right)^2 = \lambda \left( \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j}^2 x_{i,j} + K^2 - 2K \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j} x_{i,j} + \sum_{(i,j) \neq (k,l)} d_{i,j} d_{k,l} x_{i,j} x_{k,l} \right) . \tag{6}$$

**Step 4 — Combined Hamiltonian.** The total QUBO Hamiltonian $H(\mathbf{x})$ is the sum of the negated objective and the penalty terms:

$$H(\mathbf{x}) = - \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} s_{i,j} x_{i,j} + \lambda \sum_{i \in \mathcal{N}} \left( 1 - \sum_{j \in \mathcal{C}_i} x_{i,j} \right) + \lambda \left( \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j}^2 x_{i,j} + K^2 - 2K \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{C}_i} d_{i,j} x_{i,j} + \sum_{(i,j) \neq (k,l)} d \right. \tag{7}$$

**Step 5 — Q Matrix Entries and Offset.** Collecting linear and quadratic coefficients, we obtain the QUBO form $H(\mathbf{x}) = \sum_{i,j} Q_{i,j} x_{i,j} + \sum_{(i,j)<(k,l)} 2Q_{(i,j),(k,l)} x_{i,j} x_{k,l} + c$. The entries are given by:

$$Q_{i,j} = -s_{i,j} - \lambda + \lambda d_{i,j}^2 - 2\lambda K d_{i,j}, \quad \forall i \in \mathcal{N}, j \in \mathcal{C}_i, \tag{8}$$

$$Q_{(i,j),(k,l)} = \lambda d_{i,j} d_{k,l}, \quad \forall (i,j) \neq (k,l), \tag{9}$$

$$c = \lambda N + \lambda K^2 . \tag{10}$$

**Penalty Weight Justification.** Let $S_{\max} = \max_{i,j} |s_{i,j}|$. The maximum possible gain in the objective function from flipping a single variable $x_{i,j}$ is bounded by $2S_{\max}$. Violating the first constraint by one unit changes the penalty by at least $\lambda$. Violating the second constraint by one unit changes the penalty by at least $\lambda(2K + 1)$ in the worst case. To ensure that

any infeasible solution has strictly higher energy than the best feasible solution, we require $\lambda > S_{\max} + \lambda \cdot (\text{max constraint violation impact})$. A sufficient condition is $\lambda > \sum_{i,j} |s_{i,j}| + \lambda \cdot (2KD_{\max} + D_{\max}^2)$, where $D_{\max} = \max_{i,j} d_{i,j}$. In practice, choosing $\lambda > \max_{i,j} |s_{i,j}|$ ensures that the penalty cost for any constraint violation strictly exceeds the maximum achievable objective improvement, thereby forcing the optimizer to satisfy all constraints. For typical normalized scores bounded by $|s_{i,j}| < 100$, setting $\lambda = 1000$ guarantees this dominance.

**Correctness Argument.** Minimizing $H(\mathbf{x})$ over $\{0,1\}^n$ recovers the optimal solution because the penalty terms are non-negative and vanish exactly on the feasible set defined by the one-parent-set and total-edge constraints. For any feasible assignment, $H(\mathbf{x})$ reduces to the negated objective function, so the global minimum corresponds to the maximum score. For any infeasible assignment, at least one constraint is violated, incurring a positive penalty. By choosing $\lambda$ sufficiently large, the energy increase from constraint violation strictly dominates any potential reduction in the objective function, ensuring that the global minimum cannot be infeasible. Thus, the unconstrained QUBO minimum is guaranteed to be feasible and optimal for the original constrained problem.

**Illustrative Example.** Consider a small instance with $N = 3$ nodes $\{0, 1, 2\}$ and required edges $K = 2$. Suppose each node has exactly two candidate parent sets ($\mathcal{C}_i = \{0, 1\}$ for $i = 0, 1, 2$). Let the scores and sizes be: Node 0: $(s_{0,0}, d_{0,0}) = (10, 0)$, $(s_{0,1}, d_{0,1}) = (20, 1)$. Node 1: $(s_{1,0}, d_{1,0}) = (25, 1)$, $(s_{1,1}, d_{1,1}) = (5, 0)$. Node 2: $(s_{2,0}, d_{2,0}) = (30, 1)$, $(s_{2,1}, d_{2,1}) = (10, 0)$. With $\lambda = 1000$, the diagonal QUBO coefficients are:

$$Q_{0,0} = -10 - 1000 + 0 - 0 = -1010, \tag{11}$$
$$Q_{0,1} = -20 - 1000 + 1000(1)^2 - 2000(2)(1) = -4020, \tag{12}$$
$$Q_{1,0} = -25 - 1000 + 1000(1)^2 - 2000(2)(1) = -4025, \tag{13}$$
$$Q_{1,1} = -5 - 1000 + 0 - 0 = -1005, \tag{14}$$
$$Q_{2,0} = -30 - 1000 + 1000(1)^2 - 2000(2)(1) = -4030, \tag{15}$$
$$Q_{2,1} = -10 - 1000 + 0 - 0 = -1010. \tag{16}$$

The off-diagonal coefficients are $Q_{(i,j),(k,l)} = 1000 \cdot d_{i,j} d_{k,l}$. Only pairs where both candidates have $d \geq 1$ yield non-zero off-diagonals. Here, $d_{0,1} = 1, d_{1,0} = 1, d_{2,0} = 1$. Thus, $Q_{(0,1),(1,0)} = 1000$, $Q_{(0,1),(2,0)} = 1000$, and $Q_{(1,0),(2,0)} = 1000$. All other off-diagonal entries are zero. The constant offset is $c = 1000(3) + 1000(2^2) = 7000$.

The QUBO Hamiltonian is $H(\mathbf{x}) = \sum_{i,j} Q_{i,j} x_{i,j} + \sum_{(i,j)<(k,l)} 2Q_{(i,j),(k,l)} x_{i,j} x_{k,l} + 7000$. Enumerating the $2^3 = 8$ feasible assignments (one per node), we check edge counts and scores:

- $\mathbf{x} = (1, 0, 1, 0, 1, 0)$: edges $0+1+1 = 2$, score $10+25+30 = 65$. $H = -65+0+7000 = 6935$.
- $\mathbf{x} = (0, 1, 0, 1, 1, 0)$: edges $1+0+1 = 2$, score $20+5+30 = 55$. $H = -55+0+7000 = 6945$.

All other assignments violate the edge constraint and incur a penalty of at least 1000, yielding energies strictly above 7000. The minimum energy is 6935 at $\mathbf{x} = (1, 0, 1, 0, 1, 0)$, corresponding to selecting candidate 0 for all nodes. This yields exactly $K = 2$ edges and the maximum score of 65, confirming the formulation's correctness on this instance.